

Subject: Digital Forensic and Hacking Techniques	Subject Code: 316315
Semester: 06	Course: CO6K-A
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Experiment No:	12
Title of Experiment:	Establish DoS attack using TCP/ICMP flooding: a. Ping continuously a particular machine at a time from different machines and observe the machine behavior on Network. b. Write shell script for continuously flooding a Machine with ping and observe the machine behavior on Network.

❖ Practical Significance:

TCP and ICMP floods remain important because they're simple, low-cost ways for attackers to deny service, cause outages, impose costs, and test or probe defenses while defenders need pragmatic detection and layered mitigation to reduce impact.

➤ Procedures

1. Safe Lab Setup

- Create two virtual machines (VMs) on your computer using **VirtualBox or VMware**.
- Assign them IPs in the same network (e.g., 192.168.56.101 and 192.168.56.102).
- Only perform tests between these VMs.
- Install Linux (Ubuntu) on both.

2. Ping Flood (Safe Simulation)

a) Observing network behavior

You can use ping to simulate load on your own VM safely:

ping -i 0.1 192.168.56.102

- -i 0.1 → Sends a ping every 0.1 seconds (increase network load in lab only).

- Monitor the target VM using:

top

or

htop

to see CPU/network usage, or

iftop

for network monitoring.

Only do this in a controlled lab; never target public IPs.

➤ Practical Related Questions:

1. What is a DoS / DDoS attack?

Ans:

DoS (Denial of Service) and **DDoS (Distributed Denial of Service)** attacks are cyber-attacks used to **make a website, server, or network unavailable to users**.

- **DoS: A single computer** sends many requests to overload a server, causing it to slow down or crash.
- **DDoS: Many computers** (botnet) send massive requests at the same time, making the attack much stronger and harder to stop.

2. How does a TCP Flood exploit this?

Ans:

A **TCP Flood** is a type of **DoS/DDoS attack** where the attacker sends a huge number of **TCP connection requests** to a server.

Because of the **TCP three-way handshake** (SYN → SYN-ACK → ACK), the server starts creating connections for each request. If attackers send **many fake SYN requests and never complete the handshake**, the server's resources get filled.

As a result:

- The server becomes **slow or overloaded**
- **Legitimate users cannot connect**

3. Why is it illegal to perform TCP floods?

Ans:

Performing a **TCP flood** is illegal because it is a **cyber attack that intentionally disrupts or damages computer systems and networks**.

Reasons:

- **Service disruption:** It overloads servers, making websites or services unavailable to legitimate users.
- **Financial loss:** Businesses can lose money when their services go down.
- **Unauthorized interference:** It interferes with systems without permission.
- **Violation of cyber laws:** Most countries have strict **cybercrime laws** that punish such attacks.

❖ Conclusion

The experiment demonstrated how TCP/ICMP flooding can create a Denial of Service (DoS) by sending many requests to a target machine. Continuous ping requests from multiple systems or through a shell script overload the target's network resources, causing slow response, packet loss, high CPU/network usage, or temporary unavailability of services.

This practical activity highlights how even simple and low-cost attack methods can significantly impact system performance and network availability. It also emphasizes the importance of network monitoring, rate limiting, firewalls, and intrusion detection systems to detect and mitigate such attacks. Understanding these attacks helps cybersecurity professionals design stronger defense mechanisms to protect systems from service disruption.

Marks Obtained			Dated Signature of Teacher
Process Related (15)	Product Related (10)	Total (25)	